YUE YU

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EDUCATION

INDIANA UNIVERSITY (GPA: 3.9/4.0) **Bloomington, IN**
Ph.D. in Statistical Science, with Minor in Computer Science and Mathematics *Aug. 2021 – Present*

- **Core Courses:** Applied Statistical Computing, Advanced Statistical Theory I & II, etc.
- **Appointment:** Teaching Assistant for STAT S520-Introduction to Statistics (Fall 2021, Spring 2022)
- **Award:** Department of Statistics Fellowship (Summer 2022)

UNIVERSITY OF MICHIGAN (GPA: 3.7/4.0) **Ann Arbor, MI**
Master of Science in Applied Statistics *Sept. 2018 – May 2020*

- **Core Courses:** Statistical Learning I: Regression, Statistical Learning II: Multivariate Analysis, Statistical Inference, Bayesian modeling and computation, Nonparametric Statistics, etc.
- **Awards:** Rackham Graduate School Conference Travel Grant (2020); Rackham Graduate School International Student Fellowship (2019)

Master of Science in Sport Management *Sept. 2018 – May 2020*

- **Core Courses:** Financial Management for the Sport Industry, Strategic Management in Sport, Marketing Management for the Sport Industry, Philosophy of Science and Research in Kinesiology, etc.
- **Activities:** Teaching Assistant for SM313-Sports Analytics course in Winter 2020 semester; Michigan Men's Basketball Data Analytics Student Consultant in 2019-2020, provided game analysis and suggestions

SUN YAT-SEN UNIVERSITY (GPA: 3.4/4.0) **Guangzhou, China**
Bachelor of Engineering in Mechanical Engineering *Aug. 2014 – Jun. 2018*

- **Core Courses:** Advanced Mathematics 1(Highest Level) I & II, Linear Algebra, Computer Algorithmic Language, Ordinary Differential Equations, Methods of Mathematical Physics, Computational Fluid Mechanics, Theoretical Mechanics I & II, The Finite-Element Method & Programming, etc.
- **Awards:** Sun Yat-sen University Outstanding Student Scholarship (2016); Sun Yat-sen University Exchange Student Fellowship (2017)

Minor in Statistics *Aug. 2015 – Jun. 2018*

- **Core Courses:** Probability Theory & Mathematical Statistics, Mathematical Modeling Practice, Computational Methods, Operations Research, etc.

COLUMBIA UNIVERSITY (GPA: 3.7/4.0) **New York, NY**
Exchange Student Program with Concentration in Statistics *Jan. 2017 – May. 2017*

- **Courses**: Introduction to Probability & Statistics, Time Series Analysis, etc.

SELECTED PROFESSIONAL EXPERIENCES

BAIM INSTITUTE FOR CLINICAL RESEARCH **Boston, MA**
SAS Programmer I *Nov. 2020 – Jun. 2021*

- Supported the continued success and quality of biostatistics clinical research projects
- Performed SAS programming using such techniques as macro language, advanced data manipulation techniques, and statistical procedures (e.g., PROC GLM, PROC FREQ, PROC REPORT)
- Reviewed and provided feedback regarding data management plans and research manuscripts for publication

UNIVERSITY OF MICHIGAN **Ann Arbor, MI**
Research Affiliate, Department of Statistics *Aug. 2020 – Nov. 2020*

- Conducted medical machine learning projects by using and developing diverse and rigorous statistical models
- Presented findings to leadership teams and colleagues and participate in the composition of academic papers

BANK OF MONTREAL **Guangzhou, China**
Global Information and Technology Risk Management Assistant *Jan. 2018 - May 2018*

- Completed 12 main technology risk governance and supplier risk assessments and data protection projects
- Aligned objectives and bridged communications with Oracle teams to perform internet data analysis using SQL

ADIDAS **Beijing, China**
Business Analytics Assistant *Jul. 2017 – Dec. 2017*

- Tasked with collating, stratifying, and visualizing 20% of North Region sales data in Q3 & Q4 2017
- Optimized and enhanced sales prediction models by leveraging historical data and market research

RESEARCH PROJECTS

NBA Player Clustering Analysis *Mar. 2021 – Present*

Project Leader (Advised by Prof. Guanyu Hu)

- Using Bayesian Mixtures of Copulas to optimize traditional clustering methods like K-Means
- Categorizing different types of players based on clustered traditional and advanced statistics
- Revising abstract for 2023 MIT Sloan Sports Analytics Conference submission

ATUS Activities Before Primary Sleep Analysis *Jan. 2020 – Dec. 2020*

Project Leader (Advised by Prof. Galit Dunietz and Prof. Kerby Shedden)

- Supported the improvement of sleep quality by analyzing data from the American Time Use Survey and develop an algorithm to extract primary sleep from survey data
- Assessed the influence of exercise activities on sleep in different time period over the day through various statistical methods and prepare a summary of the findings for public health journal submission

Michigan Pharmacy Fraud Analysis *May 2020 – Dec. 2020*

Lab Project Member (Advised by Prof. Ji Zhu and Dr. Xianshi Yu)

- Performed feature engineering and data pre-processing tasks like dimension reduction and non-ignorable missing data imputation
- Optimized the performance of traditional ML methods by developing positive and unlabeled learning models, such as one-class SVM and isolated forest, achieving the best model with 0.88 AUC
- Delivered an engaging and comprehensive presentation on final models and visualized data for the Office of the Inspector General of the Michigan State Government

OSA Disease Spatial Analysis *Aug. 2019 - May 2020*

Project Co-Leader (Advised by Prof. Galit Dunietz)

- Evaluated and assessed data for nationwide Obstructive Sleep Apnea Syndrome (OSA) treatment
- Outlined the correlation between OSA treatment rate and ethnic group distribution within the state or Hospital Referral Region by utilizing GLM, geospatial data analysis, and visualization
- Presented findings at the SLEEP 2020 Conference (abstract accepted) and co-authored a published paper

PUBLICATION

Dunietz, G. L., **Yu, Y.**, Levine, R. S., Conceicao, A. S., Burke, J. F., Chervin, R. D., & Braley, T. J. (2020). Obstructive sleep apnea in older adults: geographic disparities in PAP treatment and adherence. Journal of clinical sleep medicine: JCSM: official publication of the American Academy of Sleep Medicine, 10.5664/jcsm.8914. Advance online publication. https://doi.org/10.5664/jcsm.8914.

CONFERENCE PRESENTATION

Yue Yu, Rivkah S. Levine, Tiffany J. Braley, James F. Burke, Ronald D. Chervin, Galit L. Dunietz. Obstructive Sleep Apnea in Older Adults: Geographic Variation in CPAP Treatment. SLEEP 2020, Oral and Poster Presentation.

ONLINE COURSES & CERTIFICATES

Stochastic Processes, National Research University Higher School of Economics on Coursera *Nov. 2020*
Data Science Professional Certificate, Harvard University on edX *Jun. 2019*
Deep Learning Specialization Certificate, deeplearning.ai on Coursera *Aug. 2019*
Machine Learning, Stanford University on Coursera *May 2019*

ADDITIONAL ACOMPLISHMENTS & HONORS

Honorable Mention, Mathematical Contest in Modeling *Jan. 2017*
Provincial 2nd Prize, China Undergraduate Mathematical Contest in Modeling *Sept. 2016*

SKILLS

Programming Languages: R (dplyr/ggplot2), Python (Tensorflow/Keras/Pandas/NumPy/matplotlib), SQL, SAS, Spark, MATLAB, C++, Fortran
Data Visualization Tools: ArcGIS, Tableau, Flourish
Software & Tools: MS Office Suite, LaTeX